🧠 Agent Skills

Turn General-Purpose AI into Domain Specialists

Geeks Club

📅 February 2026

📋 Agenda

1. 🤔 **What are Skills?** — Definition & mental model
2. 🎯 **Why use Skills?** — Benefits & use cases
3. ⚙️ **How Skills work** — Progressive disclosure & architecture
4. 🏗️ **Skill structure** — Anatomy of a SKILL.md
5. 📐 **Core principles** — Conciseness, degrees of freedom
6. ✍️ **Creating a Skill** — Step-by-step guide
7. 📦 **Patterns & examples** — Real-world Skills
8. 🚫 **Anti-patterns** — What to avoid
9. 💭 **Discussion**

🤔 What are Skills?

> Reusable, filesystem-based resources that provide AI agents with **domain-specific expertise**: workflows, context, and best practices.

Key idea:

Skills transform **general-purpose agents** into **specialists**.

Concept	Analogy
🧠 Skill	An onboarding guide for a new team member
📝 Prompt	A one-off sticky note with instructions

Skills differ from prompts — they **load on-demand** and eliminate the need to repeatedly provide the same guidance.

🎯 Why Use Skills?

Three key benefits:

- 🎓 **Specialize the agent** — Tailor capabilities for domain-specific tasks
- 🔄 **Reduce repetition** — Create once, use automatically across conversations
- 🧩 **Compose capabilities** — Combine Skills to build complex workflows

Without Skills:

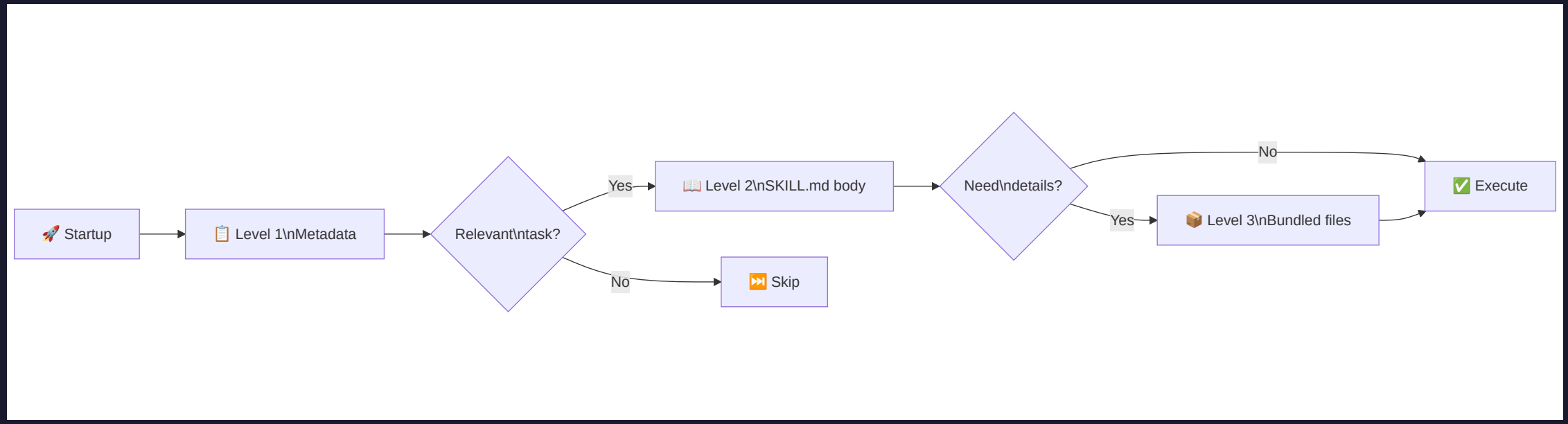
```
You: "Remember to always filter test accounts, use UTC timestamps,
     and follow our naming convention for SQLServer tables..."
     (... every single conversation)
```

With Skills:

```
You: "Analyze Q4 revenue"
Agent: (automatically loads SQLServer Skill with all your conventions)
```

⚙️ How Skills Work — Progressive Disclosure

Skills leverage a **three-level loading** strategy:



Only relevant content occupies the **context window** at any given time.

📊 Three Levels in Detail

Level	When loaded	Token cost	Content
Level 1 — Metadata	Always (startup)	~100 tokens/Skill	`name` and `description` from YAML
Level 2 — Instructions	When triggered	< 5k tokens	SKILL.md body
Level 3 — Resources	As needed	Effectively unlimited	Bundled files, scripts, schemas

Example flow:

```
User: "Extract text from this PDF"
                    ↓
Agent checks metadata: "pdf-processing" matches! → reads SKILL.md
                    ↓
SKILL.md says: "For forms, see FORMS.md" → user didn't ask about forms → skip
                    ↓
Agent uses instructions from SKILL.md to complete the task
```

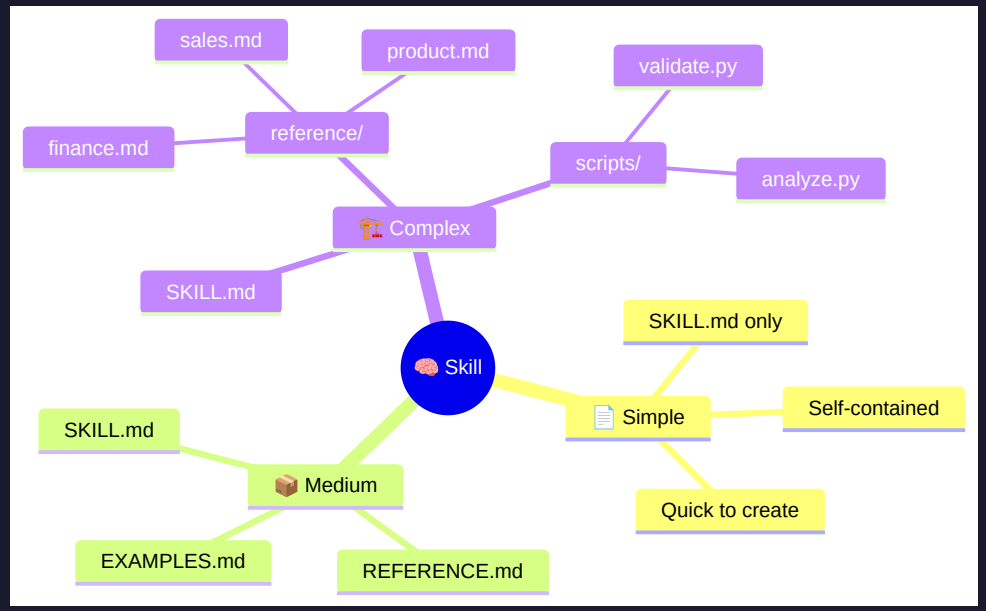
Skill Structure — Anatomy of SKILL.md

Every Skill requires a `SKILL.md` file with **YAML frontmatter**:

````
---
name: pdf-processing
description: Extract text and tables from PDF files, fill forms,
  merge documents. Use when working with PDF files or when the user
  mentions PDFs, forms, or document extraction.
---

# PDF Processing

## Quick start

Use pdfplumber to extract text from PDFs:

```python
import pdfplumber
with pdfplumber.open("document.pdf") as pdf:
    text = pdf.pages[0].extract_text()
```

## Advanced features
````

📂 Directory Structure

A Skill can be simple or complex:



Rule: Keep references one level deep from SKILL.md.

📐 Core Principle 1: Conciseness

> The **context window** is a public good. Your Skill shares it with everything else.

Default assumption: The agent is already very smart.

Only add context the agent doesn't already have. Challenge each piece:

❌ Too verbose (~150 tokens)	✅ Concise (~50 tokens)
"PDF files are a common format that contains text, images, and other content. To extract text, you'll need a library. There are many libraries available..."	"Use pdfplumber for text extraction:" + code

Ask yourself:

- 🤔 "Does the agent really **need** this explanation?"
- 🤔 "Can I **assume** the agent knows this?"

📐 Core Principle 2: Degrees of Freedom

Match **specificity** to the task's fragility:

🟢 High freedom — Multiple valid approaches

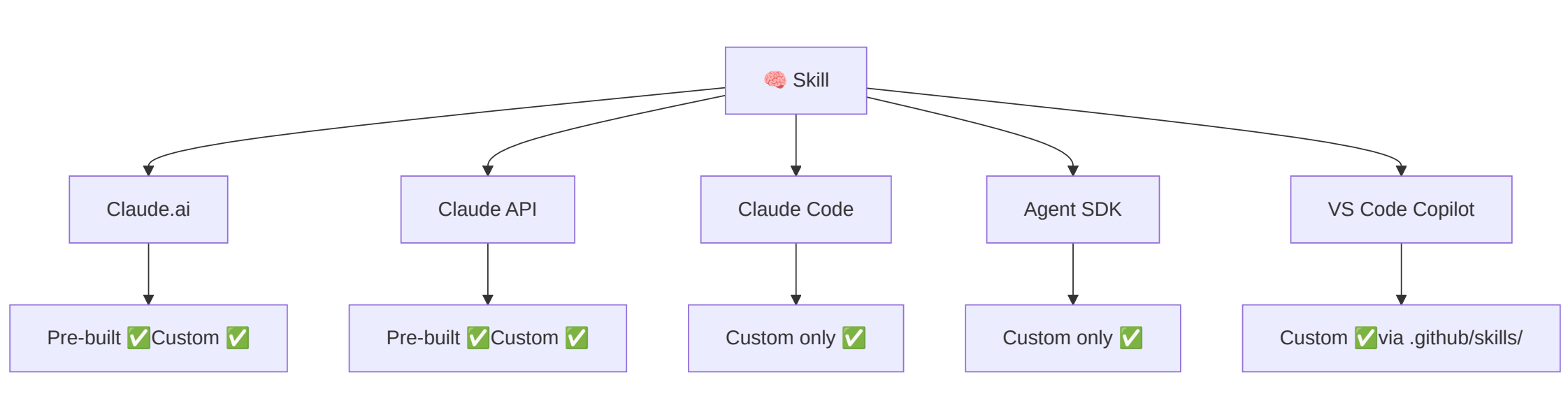
```
## Code review process
1. Analyze the code structure and organization
2. Check for potential bugs or edge cases
3. Suggest improvements for readability
```

🟡 Medium freedom — Preferred pattern with variation

```
## Generate report
Use this template and customize as needed:
def generate_report(data, format="markdown", include_charts=True):
    # Process data, generate output, optionally add visualizations
```

🔴 Low freedom — Fragile, must be exact

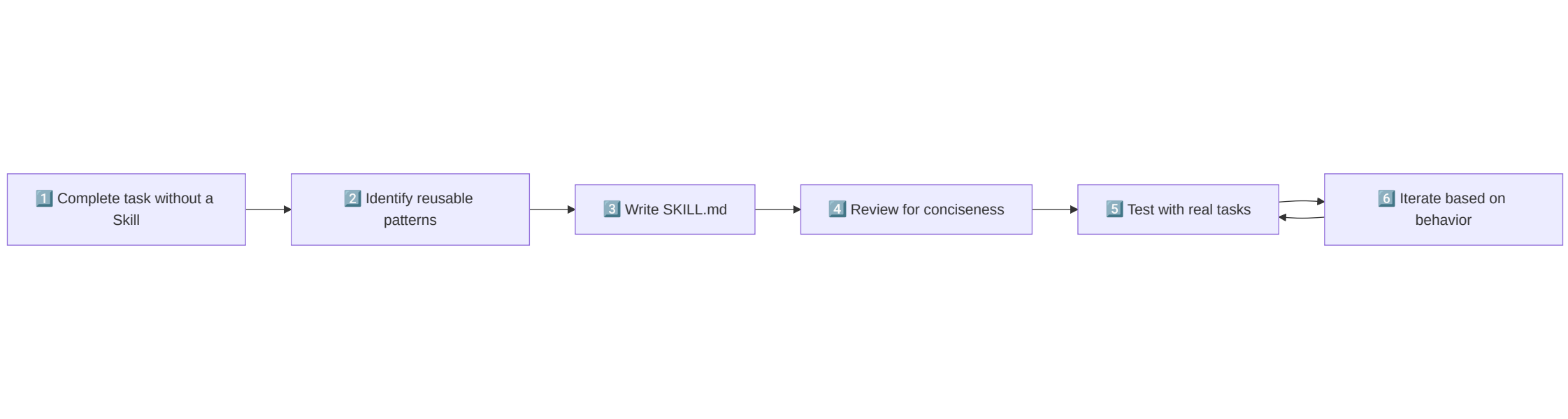
🌍 Where Skills Work



⚠️ Skills do NOT sync across surfaces

Upload/configure separately for each platform.

✍️ Creating a Skill — Step by Step



Step 1: Complete a task with normal prompting first

Step 2: Identify the reusable pattern

Step 3: Write SKILL.md

✍️ Naming & Description Best Practices

Name format: gerund form (verb + -ing)

✅ Good names	❌ Bad names
`processing-pdfs`	`helper`
`analyzing-spreadsheets`	`utils`
`managing-databases`	`documents`
`writing-documentation`	`claude-tools`

Description: what + when in third person

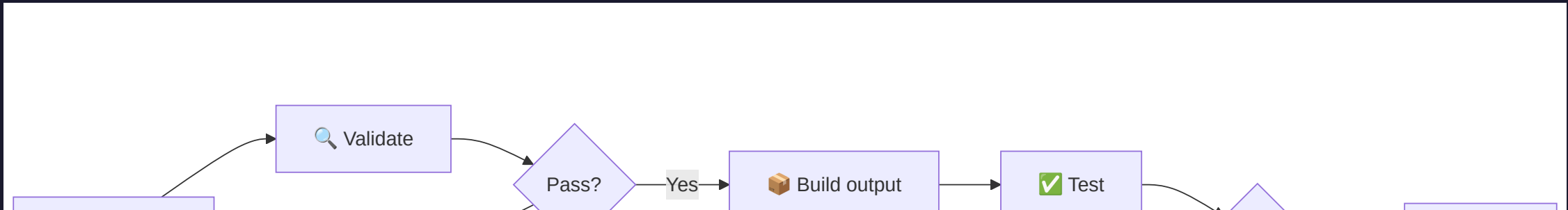
```
# ✅ Good
description: "Analyze Excel spreadsheets, create pivot tables,
  generate charts. Use when analyzing Excel files, spreadsheets,
  tabular data, or .xlsx files."
```

🔄 Workflows & Feedback Loops

Use workflows for complex tasks:

```
## PDF form filling workflow

Task Progress:
- [ ] Step 1: Analyze the form (run analyze_form.py)
- [ ] Step 2: Create field mapping (edit fields.json)
- [ ] Step 3: Validate mapping (run validate_fields.py)
- [ ] Step 4: Fill the form (run fill_form.py)
- [ ] Step 5: Verify output (run verify_output.py)
```

Implement feedback loops:



🛠️ Example: Our Presentation Skill

We use a Skill right here in this repo!

```
.github/skills/creating-geeks-club-marp-presentation/
└── SKILL.md
```

```
name: creating-geeks-club-marp-presentation
description: Creates Marp slide deck presentations for Geeks Club —
  a developer-to-developer knowledge sharing meeting. Use when the
  user asks to create, scaffold, or generate a new presentation,
  slide deck, or talk for Geeks Club.
```

What it encodes:

- 🎨 **Visual identity** — dark theme, color palette, CSS
- 📁 **Folder structure** — numbered folders, naming conventions
- 📊 **Diagram standards** — Mermaid syntax, max 8 nodes

🛠️ Example: BigQuery Analytics Skill

```
bigquery-skill/
├── SKILL.md
└── reference/
    ├── finance.md    (revenue metrics, ARR, billing)
    ├── sales.md      (opportunities, pipeline, accounts)
    ├── product.md    (API usage, features, adoption)
    └── marketing.md  (campaigns, attribution, email)
```

What SKILL.md contains:

```
## Rules
- ALWAYS filter out test accounts (account_type != 'test')
- Use UTC timestamps for all date comparisons
- Follow naming convention: {domain}_{metric}_{period}

## Quick search
Find specific metrics:
grep -i "revenue" reference/finance.md
```

🛠️ Example: Code with Utility Scripts

Pre-made scripts > Generated code

```
pdf-skill/
├── SKILL.md
└── scripts/
    ├── analyze_form.py    → Extract form fields
    ├── validate_fields.py → Check for errors
    └── fill_form.py       → Apply values to PDF
```

Why scripts are better than asking the agent to write code:

Aspect	Script	Generated code
🎯 Reliability	Battle-tested	May have bugs
💰 Token cost	Output only	Full code in context
⏱️ Speed	Instant execution	Generation time

🚫 Anti-Patterns to Avoid

❌ Over-explaining what the agent already knows

```
# Bad: "PDF files are a common format..."
# Good: "Use pdfplumber for text extraction:"
```

❌ Offering too many options

```
# Bad: "You can use pypdf, or pdfplumber, or PyMuPDF, or pdf2image..."
# Good: "Use pdfplumber. For scanned PDFs requiring OCR, use pdf2image."
```

❌ Deeply nested references

```
# Bad: SKILL.md → advanced.md → details.md → actual info
# Good: SKILL.md → advanced.md (one level deep)
```

❌ Windows-style paths

🔒 Security Considerations

> ⚠️ Use Skills **only from trusted sources** — those you created yourself or from Anthropic.

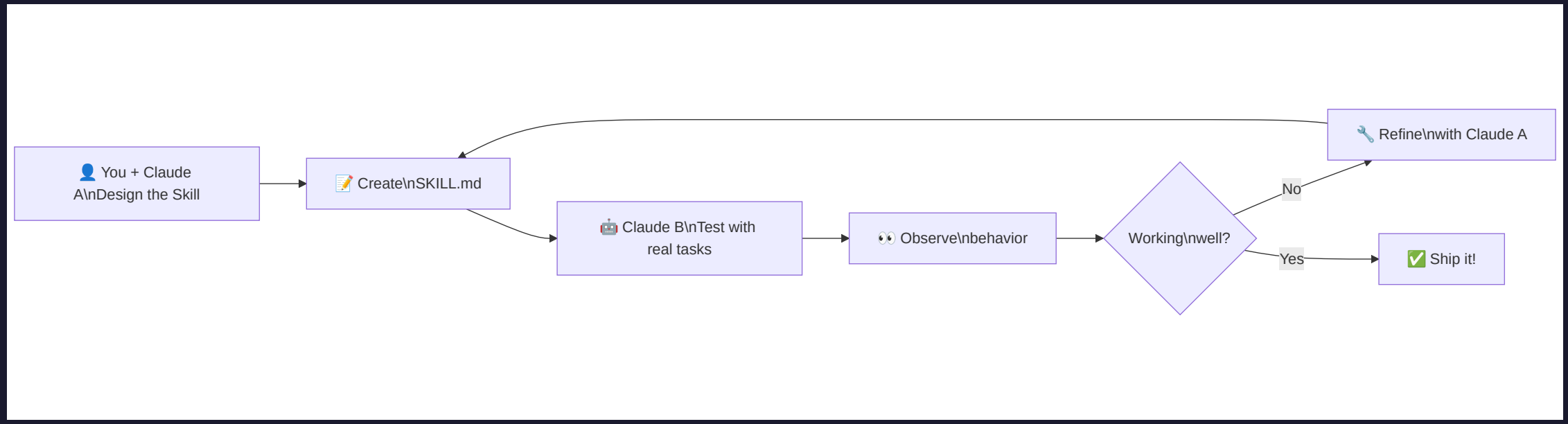
Risks of untrusted Skills:

- 🥷 **Data exfiltration** — leaking sensitive info to external systems
- 🔓 **Unauthorized access** — misusing file operations and bash commands
- 🎭 **Hidden behavior** — instructions that don't match stated purpose
- 🌐 **External dependencies** — fetched content may contain malicious instructions

Checklist before using third-party Skills:

- [ ] Review ALL files: SKILL.md, scripts, resources

🔁 Iterative Development



Two-agent pattern:

- **Claude A** (expert) — helps you design and refine the Skill
- **Claude B** (tester) — uses the Skill on real tasks, reveals gaps

Key observations to track:

✅ Checklist for Effective Skills

Core quality:

- [ ] Description is **specific** and includes trigger keywords
- [ ] SKILL.md body is **under 500 lines**
- [ ] **No time-sensitive** information
- [ ] **Consistent terminology** throughout
- [ ] **Concrete examples**, not abstract descriptions
- [ ] References are **one level deep**

Code and scripts:

- [ ] Scripts **solve problems** rather than punt to the agent
- [ ] Error handling is **explicit and helpful**

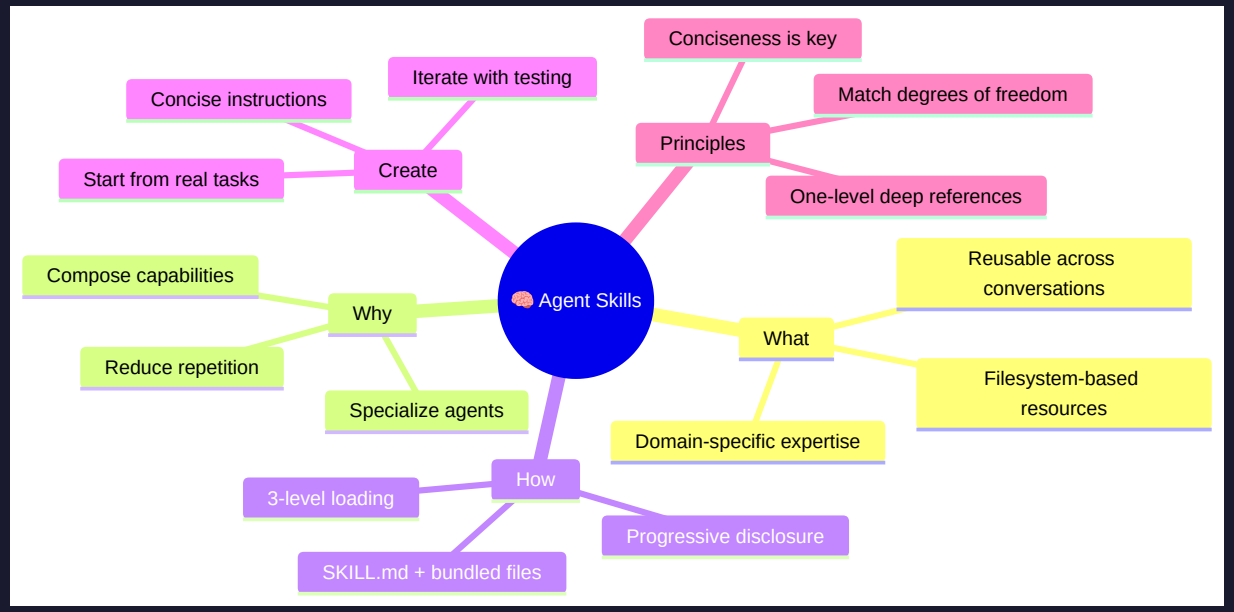
🎯 Key Takeaways

1. 🧠 **Skills = persistent domain knowledge** for AI agents
   Create once, use automatically across conversations
2. 📖 **Progressive disclosure** is the core architecture
   Only relevant content enters the context window
3. ✂️ **Conciseness is king** — the agent is already smart
   Only provide what it can't infer on its own
4. 🔄 **Iterate based on observation**, not assumptions
   Watch how the agent uses your Skill, then refine
5. 🏗️ **Start simple** — a single SKILL.md can be very powerful
   Add complexity only when needed

💭 Discussion

1. 🤔 What **repetitive context** do you provide to AI agents that could become a Skill?
2. 🏛️ How could we use Skills for **team-wide conventions** (coding standards, PR reviews, deployment procedures)?
3. 🔐 How should we handle **security review** for shared Skills in our organization?
4. 🧩 Could we build a **library of Skills** for common tasks across our projects?
5. ⚖️ Where do you draw the line between **a Skill** and **a prompt template**?

🗺️ Summary


🧠 Agent Skills
Why
Compose capabilities
Reduce repetition
Specialize agents
What
Reusable across conversations
Filesystem-based resources
Domain-specific expertise
How
3-level loading
SKILL.md + bundled files
Progressive disclosure
Create
Concise instructions
Iterate with testing
Start from real tasks
Principles
Conciseness is key
Match degrees of freedom
One-level deep references

📚 Sources

- Agent Skills Overview — Official documentation
- Skill Authoring Best Practices — Core principles & patterns
- Skills Quickstart — Getting started tutorial
- Agent Skills Cookbook — Hands-on examples
- Equipping Agents for the Real World — Engineering deep-dive blog
- Use Skills in Claude Code — Claude Code integration
- Use Skills with the API — API integration guide

🙏 Thank You!

Questions?

Geeks Club — February 2026

> "Skills transform general-purpose agents into specialists."